Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/19/2007
PI Name:	Ferrando, Arny Ph.D.		
Project Title:	Protein Turnover During Space Flight		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Galveston	**State:**	TX
Zip Code:	77550	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	95-OLMSA-01
Start Date:	10/02/1997	**End Date:**	12/31/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wolfe, Robert (University of Texas Medical Branch) Moore, Melinda (Galveston) Lane, Helen (University of Texas Medical Branch)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Determination of whole-body and muscle protein metabolism during Shuttle flight and the relationship with corresponding hormonal alterations.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	These studies will provide crucial information for the establishment of a viable ground-based model to study the perturbations of protein metabolism during space flight. In turn, this will provide the capability of testing potential countermeasures and determining nutritional requirements for future extended-duration missions. Most importantly, studying the effects of stress on an otherwise healthy population will enhance our understanding and treatment of the clinically stressed patient on Earth.
Task Progress:	Conclusion Space flight results in a down regulation of protein turnover; however, protein breakdown exceeds protein synthesis. In addition, hormonal changes which accompany space flight are conducive to a net loss of muscle and body protein. Nutritional, exercise or pharmacological interventions which promote net protein synthesis must be considered.
Bibliography Type:	Description: (Last Updated: 02/27/2004)